

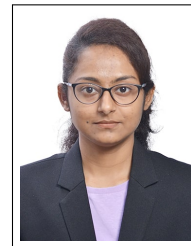
Amrita Bose

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ABOUT

Doctoral researcher working in the field of cryptography and security. At present, my research focuses on identifying and mitigating security vulnerabilities in the healthcare domain, particularly emphasizing implantable and wearable medical devices and the associated environment. During my Master's, I worked in the area of cryptanalysis using machine learning techniques. Prior to starting my academic journey at IIT Kharagpur, I gained industry experience working as an Application Developer.

RESEARCH INTERESTS

• Applied Cryptography • Cybersecurity • Cryptanalysis • Machine Learning for Security

EDUCATION

Ph.D, Computer Science and Engineering

[2025 - Present]

Indian Institute of Technology Kharagpur, West Bengal, India

Ph.D. Supervisor: Prof. Dipanwita Roy Chowdhury

MS, Computer Science and Engineering

[2023-2025]

Indian Institute of Technology Kharagpur, West Bengal, India

Thesis Title: Machine Learning Aided Cryptanalysis of Lightweight Symmetric Ciphers

Thesis Advisor: Prof. Dipanwita Roy Chowdhury

Cumulative Grade Point Average (CGPA): 9.55/10.00

B. Tech, Information Technology

[2016-2020]

MAKAUT(formerly known as WBUT), Kolkata, West Bengal

B.Tech Final Year Project: IoT Based Health Monitoring System

Degree Grade Point Average (DGPA): 9.00/10.00

PUBLICATIONS

1. **Amrita Bose**, Debranjana Pal and Dipanwita Roy Chowdhury. Deep Learning-Based Differential Distinguishers for Cryptographic Sequences. In the *Proceedings of Progress in Cryptology- INDOCRYPT 2024, 25th International Conference on Cryptology in India, Chennai, India, December 18 – 21, 2024, Proceedings Part II*, pages 114 – 133, Springer, Cham. https://doi.org/10.1007/978-3-031-80311-6_6
2. **Amrita Bose**, Debranjana Pal and Dipanwita Roy Chowdhury. Cryptographic Distinguishers Through Deep Learning for Lightweight Block Ciphers. In the *Proceedings of Applications and Techniques in Information Security, 14th International Conference, ATIS 2024, Tamil Nadu, India, November 22 – 23, 2024*, pages 47 – 63, Springer, Singapore. https://doi.org/10.1007/978-981-97-9743-1_4

CONFERENCE PRESENTATIONS

1. **INAE Technology Conclave 2026**, Bhubaneswar, Odisha, India, Presented our work, **Amrita Bose**, Anisha Mitra, Dipanwita Roy Chowdhury, “Mitigating Denial of Service Attacks on Legacy Implantable Cardioverter Defibrillators” [**Selected Best Paper & Poster Presentation**]
2. **INDOCRYPT 2024**, 25th International Conference on Cryptology in India, Chennai, Tamil Nadu, India, December 18 – 21, 2024
3. **ATIS 2024**, 14th International Conference on Applications and Techniques in Information Security, Thanjavur, Tamil Nadu, India, November 22 – 23, 2024

WORKSHOPS

1. **CCSS 2025 (CREST Crypto Summer School)** June 2025
TCG Crest, Kolkata, West Bengal, India
2. **ATTIC 2025 (Advancements in Trusted Information Computing)** March 2025
Indian Institute of Technology Kharagpur, West Bengal, India

PEER REVIEWER

1. Journal- **IEEE Transactions on Information Forensics and Security.** 2025
<https://publons.com/wos-op/review/author/3g0sx07y/>

SKILLS

• Algorithms and Data Structure • Cryptography • Machine Learning/Deep Learning • Programming Languages- C, Python

EXPERIENCE

Teaching Assistant

Department of Computer Science and Engineering, Indian Institute of Technology Kharagpur

- Programming and Data Structures Lab Spring 2026, Autumn 2025, Autumn 2024 & Autumn 2023
- B. Tech Project Session 2025-26
– Mentored a 4th year student on the project, Application of Machine Learning in Cryptanalysis. Assisted in problem formulation, methodology design and implementation
- Operating Systems Theory and Lab Spring 2024

Junior Project Officer

May 2022 - June 2024

Department of Computer Science and Engineering, Indian Institute of Technology Kharagpur

Junior Research Fellow

October 2021 - April 2022

Center of Excellence, Safety Engineering and Analytics (CoE-SEA), Indian Institute of Technology Kharagpur

Associate IT Consultant [Application Developer]

January 2021 - October 2021

ITC Infotech India Ltd., Kolkata, West Bengal, India